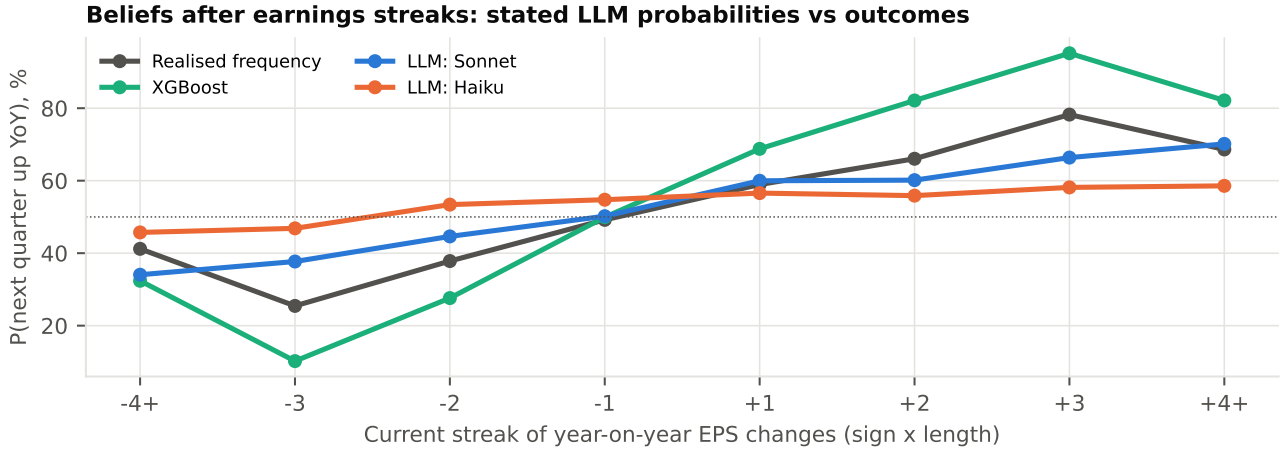
Idea 8 PoC — Do LLM forecasters inherit the gambler's fallacy?

Proof of concept, October 2026. Code: `run.py` *(model alias as argument) and* `compare.py`*.*

Question and design

LLMs are increasingly used as forecasters and as proxies for human expectations (Kim, Muhn and Nikolaev, 2024; Bybee, 2023). Do they show the gambler's fallacy, expecting a streak of earnings increases to end? Or do they show the hot hand, or neither? Each case is an *anonymised* history of 12 standalone quarterly EPS: oldest first, labelled only by fiscal quarter, rescaled by a random firm-specific constant, and without names or dates, so the model cannot recognise the firm or look ahead. The model returns a point forecast for the next quarter and p_{up}, the probability that it beats the same quarter a year earlier. There are 320 cases (FY2012–2019), stratified with 40 per bucket by the current streak of year-on-year EPS changes (-4^{+} to $+4^{+}$). Models are Claude Haiku and Claude Sonnet, called headlessly through the local `claude` CLI (20 series per call, no tools; total cost $3.65). Realised frequencies and XGBoost are computed on all 52,705 eligible cases. Under the gambler's fallacy, p_{up} would *fall* along up-streaks; under the hot hand, it would rise faster than outcomes do.



	Slope of p_{up} on up-streak length	Slope of p_{up} on down-streak length	Median abs. error (scaled units)	Direction accuracy
LLM: Haiku	0.008 (0.5)	−0.034 (−1.9)	48.6	60.3%
LLM: Sonnet	0.037 (3.9)	−0.055 (−4.9)	36.4	68.8%
Realised outcome	0.065 (2.0)	−0.060 (−1.7)	–	–
XGBoost (share of forecasts up)	0.047 (1.6)	−0.095 (−2.8)	39.8	71.6%
Seasonal random walk	–	–	45.1	–

Slopes per additional quarter of streak (t, clustered by firm), on the 320 LLM cases. Brier score of p_{up}: Haiku 0.257, Sonnet 0.207.

Findings

- **No gambler's fallacy in either model.** Neither model lowers its probability of an increase as an up-streak lengthens. Year-on-year EPS changes are persistent in the data, and both models lean, correctly, toward continuation.
- **The smaller model is conservative, not biased.** Haiku's probabilities hug 50% (46–59%) and barely respond to streaks: an underreaction pattern, like sticky expectations. Sonnet's beliefs move with the streak at about two-thirds of the realised slope, with no overreaction.
- **A capable LLM is a competitive forecaster on anonymised data.** Sonnet's point forecasts have a lower median error than XGBoost and the SRW on these cases. Its directional accuracy is slightly below XGBoost's (68.8% vs 71.6%).

Assessment and next steps

The headline hypothesis is rejected, but the PoC delivers a clean, leakage-robust protocol. It also yields an interesting dose-response: underreaction shrinks with model capability. Next steps: (i) scale to several thousand cases and more models, including open-weight ones; (ii) elicit beliefs in the *surprise* domain the paper uses ("will EPS beat the forecast?"), where the fallacy is defined; (iii) prompt variants that add a human-style narrative ("five quarters of growth in a row") to test framing; (iv) check whether LLM-implied surprises explain announcement returns better than statistical ones, as a "synthetic retail investor". Fast-moving field: best targeted at a top field journal quickly.